Introduction to GPU Programming

cuBLAS Syntax

cuBLAS Helper Functions

- cublasCreate(cublasHandle_t *handle) – initializes cublas library, must be run prior to using any other functions
- cublasDestroy() – Do this last, it releases all hardware resources
- cublasSetStream(handle, stream) – Set cublas to execute on the non-default stream
- cublasGetStream(handle, stream) – Get the stream that is currently running and determine its status

[1]

cuBLAS Level-1 Functions

Vector operations

- cublasIsamax, cublasIdamax, cublasIcamax, cublasIzamax – first index in vector with max value
- cublasIsamin, cublasIdamin, cublasIcamin, cublasIzamin – first index in vector with min value
- cublasSdot, cublasDdot, cublasCdotu, cublasCdotc, cublasZdotu, cublasZdotc – Calculates the dot product of 2 vectors
- Numerous methods for rotation, scale, etc.

Legend: s-real single precision d-real double c-complex single z-complex double

[1]

cuBLAS Level-2 Functions

Matrix-Vector Multiplication

- cublasSgbmv, cublasDgbmv, cublasCgbmv, cublasZgbmv

(cublasHandle_t handle, cublasOperation_t trans, int m, int n, int kl, int ku, const float *alpha, const float *A, int lda, const float *x, int incx, const float *beta, float *y, int incy)

$\mathbf{y}=\alpha\ op(A)\mathbf{x}+\beta\mathbf{y}$

- A slew of multiplications such as triangular, Hermitian, and symmetric kinds

[1]

Legend: s-real single precision d-real double c-complex single z-complex double

cuBLAS Level-3 Functions

Matrix-Matrix Multiplication

- cublasSgemm, cublasDgemm, cublasCgemm, cublasZgemm, cublasHgemm
- cublasCgemm3m and cublasZgemm3m – Gaussian complexity reductions
- Yet again there are triangular and Hermitian multiplication ops

Legend: s-real single precision d-real double c-complex single z-complex double

[1]

References

- [1] 2021. NVIDIA. CUDA Developer Toolkit Documentation – cuBLAS Library. Found at https://docs.nvidia.com/cuda/cublas/index.html.

JOHNS HOPKINS
UNIVERSITY

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